

VAN REENEN, South Africa

WMO: 684710

Lat: 28.3789S

Lon: 29.3853E

Elev: 5512

StdP: 12.00

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.3	31.0	8.0	10.1	55.9	11.4	12.1	54.3	26.6	58.8	23.9	59.0	7.6	0	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.5	84.2	58.1	81.4	58.8	79.2	59.2	67.7	74.3	66.3	72.9	64.9	71.6	10.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	117.0	70.7	64.3	110.9	69.4	62.9	105.5	68.0	35.8	74.4	34.4	73.1	33.2	71.5	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 21.0	18.4	16.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	22.6	68.9	3.7	1.3	19.9	69.8	17.7	70.5	15.6	71.2	12.9	72.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.0	65.7	65.5	63.8	58.8	53.9	49.7	48.9	54.0	59.5	60.8	62.4	65.2
	DBStd	8.20	4.61	4.11	4.46	5.15	4.99	5.33	6.42	6.37	7.50	7.25	6.41	5.69
	HDD50	256	0	0	0	5	22	68	100	35	15	8	3	0
	HDD65	2539	47	39	73	191	345	459	499	343	190	166	120	67
	CDD50	3533	486	434	428	270	142	60	66	158	300	343	374	472
	CDD65	341	69	53	36	6	0	0	0	1	24	37	41	74
	CDH74	2367	362	202	178	48	5	0	1	28	298	394	315	536
	CDH80	415	62	15	9	2	0	0	0	48	82	60	137	
(14) Wind	WSAvg	8.0	6.9	6.7	6.7	6.9	7.8	9.0	9.1	9.0	9.0	8.3	8.4	7.9
(15) Precipitation	PrecAvg	27.4	5.4	3.8	3.1	1.4	0.5	0.2	0.2	0.6	0.8	2.7	3.3	4.7
	PrecMax	38.7	11.5	6.5	4.9	4.0	2.6	1.0	1.8	2.7	2.9	7.1	5.6	8.9
	PrecMin	13.0	2.0	1.5	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.7	1.0
	PrecStd	6.4	2.4	1.6	1.0	0.8	0.6	0.3	0.5	0.7	0.7	1.9	1.6	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.1	82.5	81.5	79.0	74.0	70.0	71.6	77.5	84.3	86.3	85.8	88.8
		MCWB	59.9	61.9	60.5	57.3	51.7	47.4	46.3	50.8	53.3	56.3	57.7	59.4
	2%	DB	81.1	79.2	78.3	74.8	70.5	66.3	67.4	73.1	80.9	82.1	81.1	83.5
		MCWB	63.3	63.2	60.9	57.1	50.7	46.1	45.3	49.9	52.5	56.0	58.3	60.2
	5%	DB	78.0	76.4	75.8	71.6	67.7	63.2	64.2	69.7	77.3	78.6	77.6	79.5
		MCWB	63.9	63.4	61.5	56.8	50.2	45.2	44.6	48.5	51.5	55.0	58.6	61.1
	10%	DB	75.0	73.6	72.9	68.4	64.5	60.2	60.8	66.3	73.5	74.6	74.0	75.9
		MCWB	63.6	63.2	61.3	56.1	49.4	44.7	43.7	47.0	50.7	55.4	58.7	61.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.3	69.0	68.4	63.4	56.3	51.1	50.1	55.4	58.5	63.0	67.3	67.6
		MCDB	76.3	75.2	73.8	69.8	66.7	60.3	60.9	66.4	70.3	72.3	72.8	75.4
	2%	WB	67.9	67.2	65.9	60.7	54.1	49.3	47.9	53.1	56.3	60.8	64.7	65.6
		MCDB	75.0	73.2	72.3	68.2	63.4	59.0	58.7	65.3	68.8	70.4	71.2	73.5
	5%	WB	66.1	65.7	64.2	59.2	52.6	48.0	46.4	51.4	54.7	59.3	62.7	64.1
		MCDB	72.9	71.9	70.8	67.1	61.9	57.3	57.1	63.2	67.4	69.4	69.3	71.9
	10%	WB	64.6	64.3	62.5	57.8	51.1	46.5	45.1	49.7	53.4	57.9	61.1	62.7
		MCDB	71.1	70.7	69.1	65.1	60.5	55.1	55.9	61.1	66.0	67.4	68.5	70.6
(35) Mean Daily Temperature Range	5% DB	MDBR	16.5	15.6	17.0	18.8	23.4	23.1	24.1	23.7	23.8	20.3	19.2	17.7
		MCDBR	22.4	20.6	22.3	24.2	27.2	26.1	27.1	27.6	29.4	28.2	25.9	24.9
		MCWBR	9.0	8.4	9.1	10.1	12.1	12.2	12.5	11.8	10.4	9.5	8.2	8.6
	5% WB	MCDBR	18.3	17.2	18.0	19.1	22.3	21.3	23.6	23.3	23.5	21.7	19.5	19.8
		MCWBR	10.1	9.5	9.9	9.7	11.5	12.0	11.9	10.6	10.2	9.7	10.0	9.6
(40) Clear-Sky Solar Irradiance	taub		0.361	0.353	0.342	0.307	0.274	0.258	0.261	0.319	0.359	0.392	0.354	0.357
	taud		2.388	2.416	2.421	2.486	2.555	2.565	2.537	2.370	2.278	2.231	2.373	2.385
	Ebn at Noon		311	308	301	297	295	293	297	288	291	292	311	313
	Edh at Noon		41	39	36	31	26	24	26	34	41	46	41	41
(44) All-Sky Solar Radiation	RadAvg		2055	1951	1722	1445	1274	1157	1246	1462	1734	1827	2043	2112
	RadStd		162	167	116	97	65	64	57	82	94	142	134	111

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+1.03	+1.28	N/A	N/A	-0.70	-1.01	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page